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GitHub Link	https://github.com/Ganesh401697/DEEP-LEARNING

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage : 15%

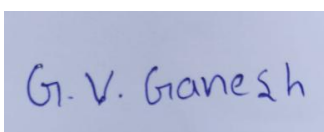
Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I G VENKATA GANESH-192425381 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: G V GANESH

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1] Unsupervised Training of Neural Networks, Autoencoders

Recommended Technique: Autoencoder

An Autoencoder is the most suitable deep learning technique for identifying abnormal ECG patterns from thousands of unlabeled ECG records. Since labeling medical data is expensive and time-consuming, autoencoders are widely used for anomaly detection in healthcare. They learn the normal structure of ECG signals without requiring class labels and detect abnormalities based on reconstruction error.

Workflow

1. **Data Collection:** Collect thousands of unlabeled ECG recordings from patients.
2. **Preprocessing:** Remove noise, normalize signal values, and segment ECG beats if necessary.
3. **Training:** Train the autoencoder using the normal ECG signals. The encoder compresses the ECG into a low-dimensional representation, while the decoder reconstructs the original signal.
4. **Feature Learning:** During training, the model automatically learns the important characteristics of normal ECG patterns.
5. **Anomaly Detection:** When a new ECG signal is given, the autoencoder reconstructs it. If the reconstruction closely matches the input, the ECG is considered normal. If the reconstruction error is high, it indicates an abnormal ECG pattern.
6. **Evaluation:** Set a reconstruction error threshold. Signals with errors greater than the threshold are classified as abnormal and sent to doctors for further diagnosis.

Advantages

- Does not require labeled ECG data.
- Learns meaningful features automatically.
- Detects unknown and rare abnormalities.
- Reduces the workload of medical experts.
- Provides high accuracy in anomaly detection.
- Suitable for large-scale hospital datasets.

Conclusion

Therefore, an Autoencoder is the best choice for this problem because it efficiently learns normal ECG patterns from unlabeled data and identifies abnormal signals through reconstruction error. This approach improves early disease detection and supports healthcare professionals in making accurate clinical decisions.

2] Autoencoders, Representation Learning

Using Autoencoders for Customer Segmentation

A retail company usually stores customer purchase information with hundreds of attributes such as product categories, purchase frequency, spending amount, payment methods, and shopping preferences. Such high-dimensional data increases computational complexity and reduces clustering performance. An Autoencoder can reduce the dimensionality of the data while preserving the important information needed for customer segmentation.

Working Process

1. **Data Collection:** Gather customer purchase records from the retail database.
2. **Data Preprocessing:** Handle missing values, normalize numerical features, and encode categorical variables.
3. **Training the Autoencoder:** Feed the customer data into the autoencoder. The encoder compresses the original features into a lower-dimensional latent representation.
4. **Representation Learning:** The compressed features retain important customer purchasing behavior while removing redundant information.
5. **Customer Segmentation:** Apply clustering algorithms such as **K-Means** or **Hierarchical Clustering** on the compressed feature vectors.
6. **Analysis:** Customers with similar purchasing patterns are grouped into different market segments.

Advantages

- Reduces dimensionality effectively.
- Removes noise and redundant features.
- Improves clustering accuracy.
- Reduces computational cost.
- Learns hidden customer behavior patterns.
- Helps in personalized marketing and product recommendations.

Conclusion

Autoencoders provide an efficient method for representation learning by generating compact feature vectors from high-dimensional customer purchase data. These learned features improve the quality of customer segmentation and enable businesses to design targeted marketing strategies, resulting in better customer satisfaction and increased sales.

3] Width and Depth of Neural Networks, Activation Functions

Recommended Architectural Changes

A shallow neural network has limited capability to learn complex facial features, resulting in poor recognition accuracy. To improve performance, the neural network architecture should be modified by increasing both the network depth and using suitable activation functions.

Recommended Improvements

1. **Increase Network Depth:** Add more hidden layers so that the network learns simple features such as edges in early layers and complex facial features in deeper layers.
2. **Increase Network Width:** Increase the number of neurons in hidden layers to improve the model's learning capacity.
3. **Use ReLU Activation:** Replace sigmoid activation in hidden layers with the Rectified Linear Unit (ReLU) because it speeds up learning and reduces the vanishing gradient problem.
4. **Use Softmax Output Layer:** Use Softmax activation in the final layer for multi-class face recognition.
5. **Batch Normalization:** Normalize layer outputs to improve training speed and stability.
6. **Dropout Regularization:** Randomly deactivate neurons during training to reduce overfitting.
7. **Data Augmentation:** Rotate, crop, flip, and brighten training images to improve model generalization.

Benefits

- Learns more complex facial features.
- Improves recognition accuracy.
- Faster convergence during training.
- Reduces overfitting.
- Better performance under different lighting conditions and facial expressions.

Conclusion

Increasing the depth and width of the neural network along with ReLU activation, Softmax output, Batch Normalization, and Dropout significantly improves facial recognition performance. These architectural changes enable the model to extract rich facial features and achieve higher accuracy in real-world applications.

4] Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Using RBMs for Movie Recommendation

A movie streaming service wants to recommend movies based on users' viewing history without using labeled data. A Restricted Boltzmann Machine (RBM) is an unsupervised neural network that learns hidden relationships between users and movies and predicts user preferences effectively.

Working Process

1. **Data Collection:** Collect user viewing history, movie ratings, and watch frequency.
2. **Create User-Movie Matrix:** Represent users as rows and movies as columns. Each value indicates whether the user watched or rated a movie.
3. **RBM Training:** The visible layer represents user preferences, while the hidden layer learns latent features such as movie genres and user interests.
4. **Feature Learning:** RBM identifies hidden patterns among users with similar viewing habits.
5. **Prediction:** Estimate the probability that a user will like unseen movies.
6. **Recommendation:** Recommend movies with the highest predicted ratings to the user.

Advantages

- Does not require labeled training data.
- Learns hidden user preferences automatically.
- Handles sparse datasets effectively.
- Generates personalized movie recommendations.
- Improves user satisfaction and engagement.

Conclusion

Restricted Boltzmann Machines are highly suitable for recommendation systems because they automatically discover hidden relationships between users and movies. By learning these latent features, RBMs provide accurate and personalized movie recommendations, improving both user experience and platform engagement.

5] Deep Learning Applications, Representation Learning

Deep Learning Solution for Defect Detection

A manufacturing company collects sensor readings every second to monitor production quality. A deep learning solution based on Representation Learning using Autoencoders can detect defective products automatically without requiring manual feature engineering.

Solution Workflow

1. **Data Collection:** Gather real-time sensor readings from manufacturing equipment.
2. **Data Preprocessing:** Remove noise, normalize sensor values, and organize the data.
3. **Model Training:** Train an autoencoder using sensor readings collected from normal products.
4. **Representation Learning:** The encoder learns compact representations of normal operating conditions.
5. **Reconstruction:** The decoder reconstructs the original sensor readings.
6. **Defect Detection:** Calculate the reconstruction error. A high reconstruction error indicates that the product differs significantly from normal patterns and is likely defective.
7. **Alert Generation:** Automatically notify quality control personnel whenever a defective product is detected.

Advantages

- Detects defects in real time.
- Learns features automatically.
- Requires minimal human intervention.
- Reduces manufacturing cost.
- Improves product quality.
- Suitable for large-scale industrial automation.

Conclusion

Representation learning with Autoencoders provides an efficient and intelligent solution for manufacturing defect detection. It continuously monitors sensor data, identifies abnormal products with high accuracy, and supports automated quality inspection, leading to improved productivity and reduced production losses.